

# One-step movement on the 11-point grid ( $n_{sd} = 2$ )

Regime 0:  $\mu = -0.0845$ ,  $\rho = 0.9900$ ,  $\sigma = 0.0243$

| i  | z      | drift   | drift/step | P(same) | P(neighbor) | P(further) |
|----|--------|---------|------------|---------|-------------|------------|
| 0  | -0.429 | 0.0034  | 0.039      | 0.953   | 0.047       | 0.000      |
| 1  | -0.341 | 0.0026  | 0.029      | 0.929   | 0.071       | 0.000      |
| 2  | -0.252 | 0.0017  | 0.019      | 0.930   | 0.070       | 0.000      |
| 3  | -0.164 | 0.0008  | 0.009      | 0.931   | 0.069       | 0.000      |
| 4  | -0.076 | -0.0001 | -0.001     | 0.931   | 0.069       | 0.000      |
| 5  | 0.013  | -0.0010 | -0.011     | 0.931   | 0.069       | 0.000      |
| 6  | 0.101  | -0.0019 | -0.021     | 0.930   | 0.070       | 0.000      |
| 7  | 0.189  | -0.0027 | -0.031     | 0.929   | 0.071       | 0.000      |
| 8  | 0.278  | -0.0036 | -0.041     | 0.928   | 0.072       | 0.000      |
| 9  | 0.366  | -0.0045 | -0.051     | 0.926   | 0.074       | 0.000      |
| 10 | 0.454  | -0.0054 | -0.061     | 0.945   | 0.055       | 0.000      |

Regime 1:  $\mu = -0.0161$ ,  $\rho = 0.7826$ ,  $\sigma = 0.0892$

| i  | z      | drift   | drift/step | P(same) | P(neighbor) | P(further) |
|----|--------|---------|------------|---------|-------------|------------|
| 0  | -0.429 | 0.0898  | 1.016      | 0.305   | 0.379       | 0.316      |
| 1  | -0.341 | 0.0706  | 0.799      | 0.284   | 0.472       | 0.244      |
| 2  | -0.252 | 0.0514  | 0.581      | 0.326   | 0.473       | 0.201      |
| 3  | -0.164 | 0.0322  | 0.364      | 0.358   | 0.480       | 0.163      |
| 4  | -0.076 | 0.0129  | 0.147      | 0.376   | 0.483       | 0.142      |
| 5  | 0.013  | -0.0063 | -0.071     | 0.379   | 0.483       | 0.138      |
| 6  | 0.101  | -0.0255 | -0.288     | 0.366   | 0.481       | 0.153      |
| 7  | 0.189  | -0.0447 | -0.506     | 0.338   | 0.476       | 0.186      |
| 8  | 0.278  | -0.0639 | -0.723     | 0.300   | 0.466       | 0.235      |
| 9  | 0.366  | -0.0831 | -0.940     | 0.254   | 0.456       | 0.290      |
| 10 | 0.454  | -0.1023 | -1.158     | 0.257   | 0.375       | 0.367      |

Regime 2:  $\mu = 0.1099$ ,  $\rho = 0.9900$ ,  $\sigma = 0.0095$

| i  | z      | drift   | drift/step | P(same) | P(neighbor) | P(further) |
|----|--------|---------|------------|---------|-------------|------------|
| 0  | -0.429 | 0.0054  | 0.061      | 1.000   | 0.000       | 0.000      |
| 1  | -0.341 | 0.0045  | 0.051      | 1.000   | 0.000       | -0.000     |
| 2  | -0.252 | 0.0036  | 0.041      | 1.000   | 0.000       | -0.000     |
| 3  | -0.164 | 0.0027  | 0.031      | 1.000   | 0.000       | -0.000     |
| 4  | -0.076 | 0.0019  | 0.021      | 1.000   | 0.000       | -0.000     |
| 5  | 0.013  | 0.0010  | 0.011      | 1.000   | 0.000       | 0.000      |
| 6  | 0.101  | 0.0001  | 0.001      | 1.000   | 0.000       | -0.000     |
| 7  | 0.189  | -0.0008 | -0.009     | 1.000   | 0.000       | 0.000      |
| 8  | 0.278  | -0.0017 | -0.019     | 1.000   | 0.000       | -0.000     |
| 9  | 0.366  | -0.0026 | -0.029     | 1.000   | 0.000       | -0.000     |
| 10 | 0.454  | -0.0034 | -0.039     | 1.000   | 0.000       | 0.000      |

$P(z' | z, s)$  before Farmer--Toda, 11-point grid

